

Assignment-3

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Subject - Statistics and Probability (MAS 2001)

Section - I

A1 The criteria of good estimator are

- (i) Unbiasedness
- (ii) Consistency
- (iii) Efficiency
- (iv) Sufficiency

A2 Write any two methods of estimation.

- (i) Point estimation
- (ii) Method of moment
- (ii) Bayesian estimation

A3 If M.L.E exists, it is the most efficient in the class of such estimator

A4 Estimate the parameter p of the binomial distribution by the method of moments.

sol

$$P(X=x) = {}^n C_x p^x (1-p)^{n-x}$$

$$E(X) = \frac{\sum_{i=1}^n x_i}{n} = n.p$$

$$= \bar{X} = n.p$$

$$\Rightarrow \hat{p} = \frac{\bar{X}}{n}$$

A5

Let T_1 and T_2 be statistics with expectations $E(T_1) = \theta_1 + \theta_2$ and $E(T_2) = \theta_1 - \theta_2$. Find unbiased estimators of θ_1 and θ_2

Sol

$$E(T_1) = \theta_1 + \theta_2 \quad \text{--- (1)}$$

$$E(T_2) = \theta_1 - \theta_2 \quad \text{--- (2)}$$

$$\text{eq. (1) + (2) } \Rightarrow$$

$$2\theta_1 = E(T_1) + E(T_2)$$

$$\theta_1 = \frac{E(T_1) + E(T_2)}{2} \quad \text{--- (3)}$$

$$\text{eq. (1) - (2) } \Rightarrow$$

$$2\theta_2 = E(T_1) - E(T_2)$$

$$\theta_2 = \frac{E(T_1) - E(T_2)}{2} \quad \text{--- (4)}$$

to estimate θ_1 and θ_2 , replace expectations $E(T_1)$, $E(T_2)$ with their statistics.

$$\hat{\theta}_1 = \frac{T_1 + T_2}{2}, \quad \hat{\theta}_2 = \frac{T_1 - T_2}{2}$$

Now there are unbiased estimator of θ_1 and θ_2 because

$$E(\hat{\theta}_1) = E\left(\frac{T_1 + T_2}{2}\right) = \frac{E(T_1) + E(T_2)}{2} = \theta_1 \quad (\text{By eq. (3)})$$

and

$$E(\hat{\theta}_2) = E\left(\frac{T_1 - T_2}{2}\right) = \frac{E(T_1) - E(T_2)}{2} = \theta_2 \quad (\text{By eq. (4)})$$

hence unbiased estimator of θ_1 and θ_2 is

$$\boxed{\hat{\theta}_1 = \frac{T_1 + T_2}{2}, \quad \hat{\theta}_2 = \frac{T_1 - T_2}{2}} \quad \underline{\underline{\text{Ans}}}$$

A6

Under what circumstances can the normal distribution be used to find confidence limits of the populations mean?

Sol

(i) The population is normally distributed, or

(ii) The sample size is large ($n > 30$), or

(iii) The population variance is known, even with a small sample size.

A-7 The interval $(\bar{x} - z_{\alpha/2} \frac{\sigma}{\sqrt{n}}, \bar{x} + z_{\alpha/2} \frac{\sigma}{\sqrt{n}})$ is known as the 95% confidence interval for μ and the 95% confidence limits are $(\bar{x} - 1.96 \frac{\sigma}{\sqrt{n}})$ and $(\bar{x} + 1.96 \frac{\sigma}{\sqrt{n}})$

A-8 Under Bayesian approach, the parameter of a distribution is a random variable

A-9 Bayes estimator is always a function of sufficient statistics.

A-10 Explain (i) Unbiasedness (ii) Consistency (iii) Efficiency (iv) Sufficiency

Sol (i) Unbiasedness: An estimator is unbiased if its expected value equals to true parameter.

$$E(\hat{\theta}) = \theta$$

(ii) Consistency: An estimator is consistent if it converges to the true parameter as sample size increases

$$\lim_{n \rightarrow \infty} \hat{\theta} = \theta$$

(iii) Efficiency: An estimator is efficient if it has the smallest variance among all estimators

(iv) Sufficiency: An estimator is said to be sufficient for a parameter θ if it contains all the information in the sample regarding the parameter.

B-1 Let $x_1, x_2, \dots, x_n$ be a random sample from $N(\mu, \sigma^2)$ population. Find sufficient estimators for μ and σ^2

Sol

$$f(x_i; \mu, \sigma^2) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x_i - \mu}{\sigma}\right)^2}$$

$$\begin{aligned} & \dots \\ & \dots \\ & \dots \end{aligned}$$

$$\begin{aligned} & -\infty < x_i < \infty \\ & -\infty < \mu < \infty \\ & \sigma > 0 \end{aligned}$$

Likelihood func. $\Rightarrow$

$$\begin{aligned} L &= f(x_1; \mu, \sigma^2) \cdot f(x_2; \mu, \sigma^2) \cdot \dots \cdot f(x_n; \mu, \sigma^2) \\ &= \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x_1 - \mu}{\sigma}\right)^2} \cdot \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x_2 - \mu}{\sigma}\right)^2} \cdot \dots \cdot \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x_n - \mu}{\sigma}\right)^2} \end{aligned}$$

$$\begin{aligned}
 L &= \frac{1}{(\sigma\sqrt{2\pi})^n} e^{-\frac{1}{2\sigma^2} [(x_1-\mu)^2 + (x_2-\mu)^2 + \dots + (x_n-\mu)^2]} \\
 &= \frac{1}{(\sigma\sqrt{2\pi})^n} e^{-\frac{1}{2\sigma^2} \sum_{i=1}^n (x_i-\mu)^2} \\
 &= \frac{1}{(\sigma\sqrt{2\pi})^n} e^{-\frac{1}{2\sigma^2} \left[\underbrace{\sum_{i=1}^n x_i^2}_{t_2(x)} + n\mu^2 - 2\mu \underbrace{\sum_{i=1}^n x_i}_{t_1(x)} \right]}
 \end{aligned}$$

from here $h(x) = 1$,

$$g(\mu, \sigma^2) (t_1(x), t_2(x)) = \frac{1}{(\sigma\sqrt{2\pi})^n} e^{-\frac{1}{2\sigma^2} [\sum x_i^2 + n\mu^2 - 2\mu \sum x_i]}$$

∴ estimator $t_1(x) = \sum_{i=1}^n x_i$ and $t_2(x) = \sum_{i=1}^n x_i^2$

are sufficient estimator for μ and σ^2 respectively. Ans

B2 find the maximum likelihood estimate for the parameter λ of a poisson distribution. On the basis of size sample size n .

Sol

By poisson distribution:

$$f(x_i) = e^{-\lambda} \lambda^{x_i}$$

Likelihood func $\Rightarrow$

$$L(\lambda) = \prod_{i=1}^n \left(\frac{e^{-\lambda} \lambda^{x_i}}{x_i!} \right)$$

$$\ln L(\lambda) = \sum_{i=1}^n [\ln e^{-\lambda} + \ln \lambda^{x_i} - \ln(x_i!)]$$

$$= \sum_{i=1}^n [-\lambda \ln e + x_i \ln \lambda - \ln(x_i!)]$$

$$= \sum_{i=1}^n [-\lambda + x_i \ln \lambda - \ln(x_i!)]$$

$$\ln L(\lambda) = -n\lambda + \sum_{i=1}^n [x_i \ln \lambda - \ln(x_i!)] \quad \text{--- (1)}$$

maximizing the likelihood func $\rightarrow$

$$\frac{\partial}{\partial \lambda} \ln L(\lambda) = -n + \sum_{i=1}^n \frac{x_i}{\lambda} \quad \text{--- ①}$$

$$\Rightarrow -n + \sum_{i=1}^n \frac{x_i}{\lambda} = 0$$

$$\Rightarrow \sum_{i=1}^n \frac{x_i}{\lambda} = n$$

$$\Rightarrow \hat{\lambda} = \frac{\sum_{i=1}^n x_i}{n} = \bar{x} \text{ (sample mean)} \quad \text{--- ②}$$

$$\frac{d^2}{(d\lambda)^2} \ln L(\lambda) = \frac{d}{d\lambda} \left(-n + \sum_{i=1}^n \frac{x_i}{\lambda} \right)$$

$$= \sum_{i=1}^n -\frac{x_i}{\lambda^2}$$

putting λ from eq. (2) $\rightarrow$

$$= - \frac{\sum_{i=1}^n x_i}{\bar{x}^2} < 0$$

ie $\hat{\lambda} = \bar{x} = \frac{\sum x_i}{n}$ is maximum likelihood estimate

B3 In random sampling from normal population $N(\mu, \sigma^2)$, find the maximum likelihood estimators for

- μ when σ^2 is known
- σ^2 when μ is known

sol ① μ when σ^2 is known:

* Likelihood function:-

$$f(x_i; \mu, \sigma^2) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x_i - \mu}{\sigma}\right)^2}$$

$$L(\mu; x_1, x_2, \dots, x_n) = \prod_{i=1}^n \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x_i - \mu}{\sigma}\right)^2}$$

$$= \frac{1}{(\sigma\sqrt{2\pi})^n} e^{-\frac{1}{2\sigma^2} \sum_{i=1}^n (x_i - \mu)^2}$$

taking Log Both sides:

$$\log L(\mu) = \log\left(\frac{1}{(\sigma\sqrt{2\pi})^n}\right)^{\frac{1}{2}n} + \log\left(e^{-\frac{1}{2\sigma^2} \sum_{i=1}^n (x_i - \mu)^2}\right)$$

$$\log L(\mu) = -\frac{n}{2} \log(2\pi\sigma^2) - \frac{1}{2\sigma^2} \sum_{i=1}^n (x_i - \mu)^2 \quad \text{--- (1)}$$

maximizing likelihood $\rightarrow$

$$\frac{d}{d\mu} \log L(\mu) = 0 - \frac{2}{2\sigma^2} \sum_{i=1}^n (x_i - \mu)(1-0) = 0$$

$$\Rightarrow \frac{1}{\sigma^2} \sum_{i=1}^n (x_i - \mu) = 0$$

$$\Rightarrow \sum_{i=1}^n x_i - n\mu = 0$$

$\Rightarrow$

$$\hat{\mu} = \frac{1}{n} \sum_{i=1}^n x_i \quad \text{--- (2)}$$

$$\frac{d^2}{d\mu^2} \log L(\mu) = -\frac{1}{\sigma^2} \sum_{i=1}^n (1)$$

$$= -\frac{n}{\sigma^2} < 0$$

(ie) MLE $\hat{\mu} = \frac{1}{n} \sum_{i=1}^n x_i$

(ii) σ^2 when μ is known: -

$$f(x_i, \mu, \sigma^2) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x_i - \mu}{\sigma}\right)^2}$$

$$L(\sigma^2, x_1, x_2, \dots, x_n) = \left(\frac{1}{\sqrt{2\pi\sigma^2}}\right)^n e^{-\frac{1}{2\sigma^2} \sum_{i=1}^n (x_i - \mu)^2}$$

taking log both side:-

$$\log L(\sigma^2) = -\frac{n}{2} \log(2\pi\sigma^2) - \frac{1}{2\sigma^2} \sum_{i=1}^n (x_i - \mu)^2$$

maximizing likelihood func -

$$\frac{d}{d\sigma^2} \log L(\sigma^2) = -\frac{n \cdot 2\pi(1)}{2 \cdot 2\pi\sigma^2} + \frac{1}{2\sigma^4} \sum_{i=1}^n (x_i - \mu)^2 \quad \text{--- (3)}$$

set it to zero

$$-\frac{n}{2\sigma^2} + \frac{1}{2\sigma^4} \sum_{i=1}^n (x_i - \mu)^2 = 0$$

$$\frac{1}{2\sigma^4} \sum_{i=1}^n (x_i - \mu)^2 = \frac{n}{2\sigma^2}$$

$$n\sigma^2 = \sum_{i=1}^n (x_i - \mu)^2$$

$$\hat{\sigma}^2 = \frac{1}{n} \sum_{i=1}^n (x_i - \mu)^2 \quad \text{--- (4)}$$

$$\frac{d^2}{d(\sigma^2)^2} \log L(\sigma^2) = \frac{n}{2\sigma^4} - \frac{1}{2\sigma^6} \sum_{i=1}^n (x_i - \mu)^2$$

putting σ^2 from eq. (4)

$$= \frac{n}{2\left(\frac{1}{n} \sum_{i=1}^n (x_i - \mu)^2\right)^2} - \frac{1}{2\left(\frac{1}{n} \sum_{i=1}^n (x_i - \mu)^2\right)^3} \sum_{i=1}^n (x_i - \mu)^2 \leq 0$$

ie Maximum likelihood estimate of σ^2 is

$$\Rightarrow \hat{\sigma}^2 = \frac{1}{n} \sum_{i=1}^n (x_i - \mu)^2$$

B4 A Quality Control expert is required to estimate the mean thickness of aluminium sheets used in the production of airframes. A sample of 100 sheets reveals a mean of 0.048 inches with a standard deviation of 0.01 inches. Construct the 99 per cent confidence interval.

Sol

$$\begin{aligned}\text{given } n &= 100 \\ \bar{x} &= 0.048 \\ \sigma &= 0.01\end{aligned}$$

interval $\Rightarrow$

$$P(-z_{\alpha/2} < Z < z_{\alpha/2}) = \gamma$$

$$P(-z_{\alpha/2} < Z < z_{\alpha/2}) = 0.99$$

$$2P(0 < Z < z_{\alpha/2}) = 0.99$$

$$P(0 < Z < z_{\alpha/2}) = 0.495$$

$$z_{\alpha/2} = 2.57$$

interval $\Rightarrow$

$$\left(\bar{x} - z_{\alpha/2} \frac{\sigma}{\sqrt{n}}, \bar{x} + z_{\alpha/2} \frac{\sigma}{\sqrt{n}} \right)$$

$$\Rightarrow \left(0.048 - \frac{2.57(0.01)}{\sqrt{100}}, 0.048 + \frac{2.57(0.01)}{\sqrt{100}} \right)$$

$$\Rightarrow (0.0454, 0.0505) \text{ Ans}$$

B5 Show that in estimating for the parameter in the Poisson distribution with general term $\frac{e^{-\mu} \mu^x}{x!}$, $\bar{x}$ is sufficient for μ .

$$f(x; \mu) = \frac{\mu^x e^{-\mu}}{x!}; \quad x=0, 1, 2, \dots$$

$$L(\mu; x) = g(T(x), \mu) \cdot h(x)$$

$$L(\mu; x_1, x_2, \dots, x_n) = \prod_{i=1}^n \frac{\mu^{x_i} e^{-\mu}}{x_i!}$$

$$= \left(\prod_{i=1}^n \frac{\mu^{x_i}}{x_i!} \right) e^{-n\mu}$$

$$= \frac{\mu^{\sum_{i=1}^n x_i}}{\prod_{i=1}^n x_i!} e^{-n\mu}$$

$$= \frac{\left(\mu^{\sum_{i=1}^n x_i} e^{-n\mu} \right)}{\prod_{i=1}^n x_i!}$$

$$= \underbrace{\left(\mu^{\sum_{i=1}^n x_i} e^{-n\mu} \right)}_{g(T(x), \mu)} \cdot \underbrace{\frac{1}{\prod_{i=1}^n x_i!}}_{h(x)}$$

Since likelihood func can be factorized as the product of func of $T(x)$ and μ and a function of data alone,

We conclude that the statistic $T(x) = \sum_{i=1}^n x_i$ is sufficient for μ .

Q1 Let $x_1 = 0.7$, $x_2 = 1.5$, $x_3 = 0.3$, $x_4 = 0.5$, $x_5 = 3.4$
be a sample of size 5 from a distribution having
pdf $f(x, \theta) = \theta e^{-\theta x}$, $x > 0$, $\theta > 0$

find the MLE of θ

Sol

$$f(x, \theta) = \theta e^{-\theta x}$$

Likelihood function -

$$\begin{aligned} L(\theta) &= \prod_{i=1}^5 (\theta e^{-\theta x_i}) \\ &= \theta e^{-\theta x_1} \cdot \theta e^{-\theta x_2} \cdot \theta e^{-\theta x_3} \cdot \theta e^{-\theta x_4} \cdot \theta e^{-\theta x_5} \\ &= \theta^5 e^{-\theta \sum_{i=1}^5 x_i} \\ &= \theta^5 e^{-\theta(0.7 + 1.5 + 0.3 + 0.5 + 3.4)} \end{aligned}$$

$$L(\theta) = \theta^5 e^{-\theta(6.4)}$$

taking log both side

$$\log L(\theta) = 5 \log \theta - \theta(6.4)$$

maximizing Likelihood func -

$$\frac{d(\log L(\theta))}{d\theta} = \frac{5}{\theta} - 6.4 \quad \text{--- (1)}$$

equating to zero -

$$\frac{5}{\theta} - 6.4 = 0$$

$$\hat{\theta} = \frac{5}{6.4} = 0.78125$$

$$\frac{d^2}{d\theta^2} (\log L(\theta)) = -\frac{5}{\theta^2} < 0$$

ie maximum likelihood

estimation $\Rightarrow \hat{\theta} = 0.78125$ Ans

Q2 Let $x_1, x_2, \dots, x_n$ be a random sample of n observations from a population having pdf $f(x, \theta) = \theta e^{-\theta x}$, $x > 0$.

The prior distribution of θ is $f_1(\theta) = \frac{1}{\theta}$, $0 < \theta < \infty$.

Find the Bayes estimator of θ .

Sol

* prior distribution $\Rightarrow$

$$f_1(\theta) = \frac{1}{\theta} \quad \text{--- (1)}$$

* Likelihood distribution $\Rightarrow$

$$L(\theta) = \prod_{i=1}^n \theta e^{-\theta x_i}$$

$$L(\theta) = \theta^n e^{-\theta \sum_{i=1}^n x_i} \quad \text{--- (2)}$$

* ~~posterior~~ posterior $\rightarrow$

$$f(\theta/x) \propto f_1(\theta) \cdot L(\theta)$$

$$\propto \frac{1}{\theta} \cdot \theta^n e^{-\theta \sum_{i=1}^n x_i}$$

$$\propto \theta^{n-1} e^{-\theta \sum_{i=1}^n x_i}$$

Similar to gamma distribution general form $\rightarrow$

$$\text{Gamma}(\alpha, \beta) \propto \theta^{\alpha-1} e^{-\beta\theta}, \quad \alpha > 0, \beta > 0$$

Comparing with ~~posterior~~ ^{posterior} $\rightarrow$

$$\alpha = n, \quad \beta = \sum_{i=1}^n x_i$$

Bayes estimator $\Rightarrow$

$$\hat{\theta}_{\text{Bayes}} = E(\theta/x) = \frac{\alpha}{\beta} \quad (\text{mean of Gamma distribution})$$

$$\hat{\theta} = \frac{n}{\sum_{i=1}^n x_i} \quad \underline{\text{Ans}}$$

C3 A random sample of size 10 was drawn from a normal population with an unknown mean and a variance of 44.1 (inch)^2 . If the observations are (in inches): 65, 71, 80, 76, 78, 82, 68, 72, 65 and 81, obtain the 95% confidence interval for the population mean.
(tabulated value = 2.262)

Sol given, $n=10$,
 $\sigma^2 = 44.1 \text{ inch}^2$
 $\Rightarrow \sigma = 6.64 \text{ inch}$
 $t_{\alpha/2} = 2.262$

$$\bar{x} = \frac{65 + 71 + 80 + 76 + 78 + 82 + 68 + 72 + 65 + 81}{10}$$

$$\bar{x} = \frac{738}{10}$$

$$\bar{x} = 7.38$$

Interval when population mean is unknown $\Rightarrow$

$$(\bar{x} - t_{\alpha/2} \frac{\sigma}{\sqrt{n}}, \bar{x} + t_{\alpha/2} \frac{\sigma}{\sqrt{n}})$$

$$\Rightarrow \left(7.38 - 2.262 \times \frac{6.64}{\sqrt{10}}, 7.38 + 2.262 \times \frac{6.64}{\sqrt{10}} \right)$$

$$\Rightarrow (2.63, 12.13) \text{ } \underline{\underline{\text{Ans}}}$$

Q-C4 For a $N(\mu, \sigma^2)$ population with variance equal to 9, Consider a 95% confidence interval for the parametric function $r(\theta) = 2\mu + 7$, based on the random sample size 25, giving a sample mean of 30. (tabulated value = 1.96)

Sol given $\sigma^2 = 9 \Rightarrow \sigma = 3$
 $n = 25$
 $\bar{x} = 30$

Interval:-

$$P(-z_{\alpha/2} < Z < z_{\alpha/2}) = 0.95$$

$$P(-z_{\alpha/2} < Z < z_{\alpha/2}) = 0.95$$

$$2P(0 < Z < z_{\alpha/2}) = 0.95$$

$$P(0 < Z < z_{\alpha/2}) = 0.475$$

$$z_{\alpha/2} = 1.96$$

interval $\Rightarrow$

$$\left(\bar{x} - z_{\alpha/2} \frac{\sigma}{\sqrt{n}}, \bar{x} + z_{\alpha/2} \frac{\sigma}{\sqrt{n}} \right)$$

$$\Rightarrow \left(30 - \frac{1.96 \times 3}{\sqrt{25}}, 30 + \frac{1.96 \times 3}{\sqrt{25}} \right)$$

$$\Rightarrow (28.824, 31.176)$$

applying transformation $\Rightarrow$

$$r(\theta) = 2\mu + 7$$

Confidence interval $\Rightarrow$

$$\Rightarrow (2(28.824) + 7, 2(31.176) + 7)$$

$$\Rightarrow (64.648, 69.352) \quad \underline{\text{Ans}}$$

C5 Let X be a binomial random variable with parameters n and p where $f_1(p) = 1$, $0 < p < 1$.

Find the Bayes estimator of p

Sol * prior distribution $\rightarrow$

$$f_1(p) = 1 \quad \text{--- (1)}$$

* Likelihood distribution $\rightarrow$

$$L(p) = {}^n C_x p^x (1-p)^{n-x}$$

* Posterior $\rightarrow$

$$\text{Posterior}(p|X=x) \propto ({}^n C_x p^x (1-p)^{n-x})(1)$$

Since ${}^n C_x$ is constant wrt p , can be dropped

$$\text{Posterior}(p|x=x) \propto p^x(1-p)^{n-x}$$

This is Beta distribution:

$$\text{Posterior}(p|x=x) \sim \text{Beta}(\alpha+1, n-x+1)$$

Bayes estimator is expected value of ^{Beta(α, β)} this function $\Rightarrow$

$$E(p|x=x) = \frac{\alpha}{\alpha+\beta}$$

$$E(p|x=x) = \frac{\alpha+1}{(\alpha+1)+(n-x+1)} = \frac{\alpha+1}{n+2}$$

hence Bayes estimator of $p \Rightarrow$

$$\hat{p}_{\text{Bayes}} = \frac{\alpha+1}{n+2} \quad \underline{\text{value}}$$

D2 X_1, X_2 and X_3 is a random sample of size 3 from a population with mean value μ and variance σ^2 , T_1, T_2, T_3 are the estimators used to estimate mean value μ , where

$$T_1 = X_1 + X_2 - X_3$$

$$T_2 = 2X_1 + 3X_3 - 4X_2 \text{ and}$$

$$T_3 = \frac{1}{3}(\lambda X_1 + X_2 + X_3)$$

(i) Are T_1 and T_2 unbiased estimators?

(ii) find the value of λ such that T_3 is unbiased estimators for μ

(iii) With this value of λ , is T_3 a consistent estimator

(iv) Which is the best estimator?



(i) to show, $E(T_1) = \mu$ and $E(T_2) = \mu$

$$\times E(T_1) = E(X_1 + X_2 - X_3)$$

$$= E(X_1) + E(X_2) - E(X_3)$$

$$\therefore E(T_1) = \mu + \mu - \mu$$

$$\boxed{E(T_1) = \mu}$$

$$\begin{aligned}
 * E(T_2) &= E(2X_1 + 3X_3 - 4X_2) \\
 &= 2E(X_1) + 3E(X_3) - 4E(X_2) \\
 &= 2\mu + 3\mu - 4\mu
 \end{aligned}$$

$$E(T_2) = \mu$$

$$\therefore E(T_1) = E(T_2) = \mu$$

ie T_1 and T_2 are unbiased $\Rightarrow$ Ans

2 (ii) T_2 is unbiased if

$$E(T_2) = \mu$$

$$\Rightarrow \frac{1}{3} [\lambda E(X_1) + E(X_2) + E(X_3)] = \mu$$

$$\Rightarrow \frac{1}{3} [\lambda\mu + \mu + \mu] = \mu$$

$$\Rightarrow \lambda\mu + 2\mu = 3\mu$$

$$\Rightarrow \lambda\mu = \mu$$

$$\Rightarrow \boxed{\lambda = 1} \Rightarrow \text{Ans}$$

So for $\lambda = 1$, T_2 is unbiased.

(iii) T_3 consistent estimator?

ie $\text{var} \rightarrow 0$ if sample size $\rightarrow \infty$

$$\text{as } T_3 = \frac{1}{3} (X_1 + X_2 + X_3)$$

$$T_3 = \bar{X}$$

$$\therefore \text{Var}(\bar{X}) = \frac{\sigma}{n}$$

$$n \rightarrow \infty, \lim_{n \rightarrow \infty} \frac{\sigma}{n} = 0$$

$$\text{ie } \text{var}(\bar{X}) \rightarrow 0$$

ie T_3 is consistent $\Rightarrow$ Ans

(iv)

$$\text{Var}(T_1) = \text{Var}(X_1 + X_2 - X_3)$$

$$= \text{Var}(X_1) + \text{Var}(X_2) + \text{Var}(X_3)$$

$$= \boxed{3\sigma^2}$$

$$\text{Var}(T_2) = \text{Var}(2X_1 + 3X_2 - 4X_3)$$

$$= 4\text{Var}(X_1) + 9\text{Var}(X_2) + 16\text{Var}(X_3)$$

$$= 4\sigma^2 + 9\sigma^2 + 16\sigma^2$$

$$= 29\sigma^2$$

$$\text{Var}(T_3) = \text{Var}\left(\frac{1}{3}X_1 + \frac{1}{3}X_2 + \frac{1}{3}X_3\right)$$

$$= \frac{1}{9}\text{Var}(X_1) + \frac{1}{9}\text{Var}(X_2) + \frac{1}{9}\text{Var}(X_3)$$

$$= \frac{1}{9}\sigma^2 + \frac{1}{9}\sigma^2 + \frac{1}{9}\sigma^2$$

$$= \frac{1}{3}\sigma^2$$

the one with lowest Var is Best estimator

hence T_3 is Best estimator $\Rightarrow$ Ans

Q. 23 A random sample of 50 sales invoices was taken from a large population of sales invoices. The average value was found to be Rs 2000 with a standard deviation of Rs 540. Find a 90 percent confidence interval for the true mean value of all the sales.
(tabulated value = 1.645)

Sol

90% Confidence interval $\Rightarrow$

$$\left(\bar{x} - Z_{\alpha/2} \frac{\sigma}{\sqrt{n}}, \bar{x} + Z_{\alpha/2} \frac{\sigma}{\sqrt{n}} \right)$$

given $\bar{x} = 2000$

$\sigma = 540$

$n = 50$

$\times$ interval $\Rightarrow$

$$\left(2000 - 1.645 \frac{540}{\sqrt{50}}, 2000 + 1.645 \frac{540}{\sqrt{50}} \right)$$

$$\Rightarrow (1874.3754, 2125.6246) \Rightarrow \underline{\underline{\text{Ans}}}$$

Q4 The personal department of an organization would like to estimate the family dental expenses of its employees to determine the feasibility of providing a dental insurance plan. A random sample of 10 employees reveals the following family dental expenses (in thousands Rs) in the previous year: 11, 37, 25, 62, 51, 21, 18, 43, 32, 20. Set up a 99 percent confidence interval for the avg family dental expenses for the employees of this organization.
(tabulated value $t^* = 1.833$)

Sol

$$\bar{x} = \frac{11 + 37 + 25 + 62 + 51 + 21 + 18 + 43 + 32 + 20}{10}$$

$$\bar{x} = 32$$

given $n = 10$

$$s = \sqrt{\frac{\sum (x_i - \bar{x})^2}{n-1}} = \sqrt{\frac{441 + 25 + 49 + 900 + 361 + 121 + 196 + 121 + 0 + 144}{9}}$$

$$s = \sqrt{262}$$

$$s = 16.186$$

99 % Confidence interval :-

$$\left(\bar{x} - t_{\alpha/2} \frac{s}{\sqrt{n}}, \bar{x} + t_{\alpha/2} \frac{s}{\sqrt{n}} \right)$$

$$\Rightarrow \left(32 - \frac{(16.186)(1.833)}{\sqrt{10}}, 32 + \frac{(1.833)(16.186)}{\sqrt{10}} \right)$$

$$\Rightarrow \boxed{(22.61, 41.38)} \text{ Ans}$$

Q5

Survey conducted by a shopping mall group showed that a family in a metro-city spends an average of Rs 500 on cloths every month. Suppose a sample of 81 families resulted in a sample mean of Rs 540 per month and a sample standard deviation of Rs 150, develop a 99 per cent per month by a confidence interval of the mean amount spent per month by a family.

Sol

Confidence interval:

$$P(-Z_{\alpha/2} < Z < Z_{\alpha/2}) = \gamma$$

$$P(-Z_{\alpha/2} < Z < Z_{\alpha/2}) = 0.95$$

$$2P(0 < Z < Z_{\alpha/2}) = 0.95$$

$$P(0 < Z < Z_{\alpha/2}) = 0.475$$

$$Z_{\alpha/2} = 1.96$$

Interval:-

$$\left(\bar{X} - Z_{\alpha/2} \frac{\sigma}{\sqrt{n}}, \bar{X} + Z_{\alpha/2} \frac{\sigma}{\sqrt{n}} \right)$$

given

Sample mean $\bar{X} = 540$

Sample standard deviation = ~~15~~ $\sigma = 150$

Sample size = $n = 81$

interval:-

$$\left(540 - \frac{1.96(150)}{\sqrt{81}}, 540 + \frac{1.96(150)}{\sqrt{81}} \right)$$

$$\Rightarrow (507.333, 572.666) \quad \underline{\text{Ans}}$$

DL For the Poisson distribution $p(x) = P(X=x) =$

$$\frac{1}{2} \frac{e^{-m_1} m_1^x}{x!} + \frac{1}{2} \frac{e^{-m_2} m_2^x}{x!}; \quad x = 0, 1, 2, \dots \quad \text{Show that the}$$

estimates for m_1 and m_2 by the methods of moments

are: $\mu'_1 \pm \sqrt{\mu'_2 - \mu'_1 - \mu_1'^2}$

Sol

$$p(x) = P(X=x) = \frac{1}{2} \frac{e^{-m_1} m_1^x}{x!} + \frac{1}{2} e^{-m_2} m_2^x$$

~~Since~~ Since $E(X)$ of Poisson distribution is λ parameter,
So here $\Rightarrow$

$$E(X) = \frac{m_1}{2} + \frac{m_2}{2}$$

$$E(X) = \frac{m_1 + m_2}{2}$$

$$\boxed{\mu'_1 = \frac{m_1 + m_2}{2}} \quad \text{--- (1)}$$

$$\Rightarrow m_2 = 2\mu'_1 - m_1 \quad \text{--- (2)}$$

Now

$$E(X^2) = \text{Var}(X) + E(X)^2$$

$$= \frac{m_1 + m_1^2}{2} + \frac{m_2 + m_2^2}{2}$$

$$2\mu'_2 = m_1 + m_1^2 + m_2 + m_2^2 \quad \text{--- (3)}$$

By equation (2), put value of m_2 in eqn (3)

$$2\mu'_2 = m_1 + m_1^2 + 2\mu'_1 - m_1 + 4\mu_1'^2 + m_1^2 - 4\mu_1' m_1$$

$$2\mu'_2 = 2m_1^2 + 2\mu'_1 + 4\mu_1'^2 - 4\mu_1' m_1$$

$$2m_1^2 + (-4\mu_1')m_1 + (2\mu'_1 + 4\mu_1'^2 - 4\mu_2') = 0$$

$$m_1 = \frac{4\mu_1' \pm \sqrt{16\mu_1'^2 - (4)(2)(2\mu'_1 + 4\mu_1'^2 - 2\mu_2')}}{4}$$

$$m_1 = \mu_1' \pm \sqrt{\mu_2' - (\mu_1' + 2\mu_1'^2 - \mu_2')}$$

$$\boxed{\hat{m}_1 = \mu_1' \pm \sqrt{\mu_2' - \mu_1' - \mu_1'^2}}$$

$$\text{let } m_1 = \mu_1' + \sqrt{\mu_2' - \mu_1' - \mu_1'^2}$$

So putting in (2)

$$m_2 = 2\mu_1' - \mu_1' - \sqrt{\mu_2' - \mu_1' - \mu_1'^2}$$

$$\hat{m}_2 = \mu_1' - \sqrt{\mu_2' - \mu_1' - \mu_1'^2}$$

hence estimates of m_1 and m_2 by method of moment

are $\left\{ \mu_1' \pm \sqrt{\mu_2' - \mu_1' - \mu_1'^2} \right\}$ Ans